Multiple Recovery Paths in a Lesioned Transformer

Internal Rerouting and Readout Remapping

Lorenzo Cannata | Uri Hasson

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Motivation

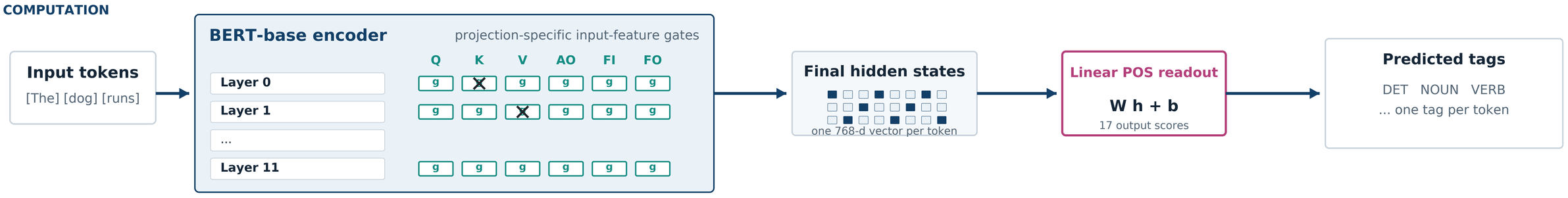
Using BERT as a frozen encoder and a linear part-of-speech (POS) readout as a task classifier, we first use gate masking to learn a sparse core subnetwork sufficient for POS classification. We then ablate this subnetwork to produce a large drop in performance. Our main question is whether recovery can be achieved by reselecting surviving projection-specific gates, remapping a learned readout, or both.

CENTRAL FINDING

Recovery of performance does not require recovery of the original representation.

Experimental framework

COMPUTATION



EXPERIMENTAL PROGRESSION

A Learn the task-sufficient core

Train projection-specific hard-concrete gates and the linear POS readout while pretrained encoder weights remain fixed.

Gates	TRAIN
Readout	TRAIN
Encoder weights	FIXED

The sparse configuration reaches 94.98% primary-run POS accuracy.

B Apply a permanent lesion

Mask scores forced to -12; effective gates are zero and gradients are blocked.

74 Q/K/V 64-coordinate input groups
5 attention-output gate tensors
9 FFN-intermediate + 4 FFN-output tensors

No recovery pathway can reactivate these entries.

C Recovery trainability

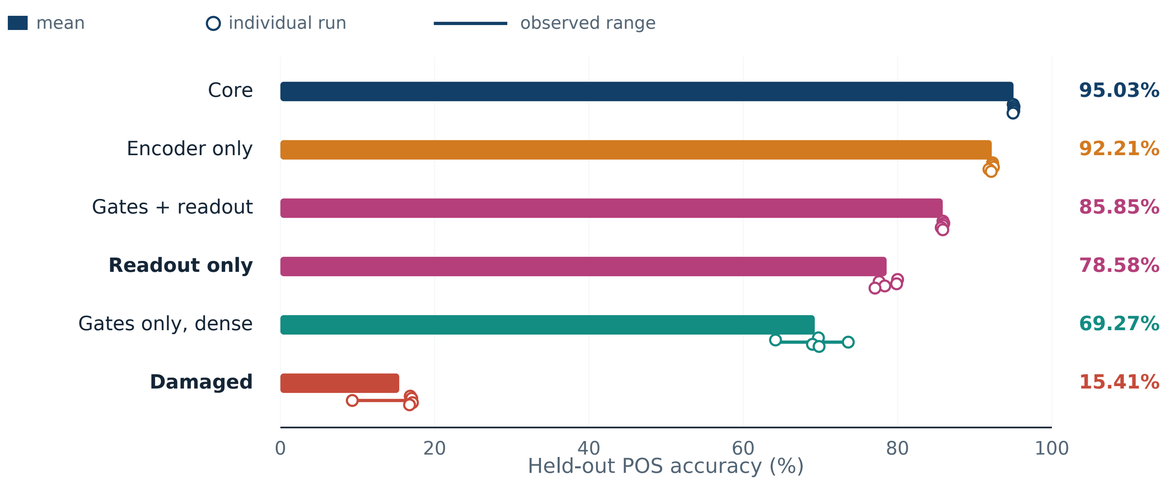
Condition	Gates	Readout	Encoder	Lesion
Gates only	TRAIN	FIXED	FIXED	OFF
Readout only	FIXED	TRAIN	FIXED	OFF
Gates + readout	TRAIN	TRAIN	FIXED	OFF
Encoder only	FIXED	FIXED	TRAIN	OFF

SETUP	bert-base-uncased | UD English-EWT | 17 POS tags | 25,147 development tokens
LESION	74 Q/K/V groups + 18 dense projection-mask tensors; 37,248 gate entries clamped

Schematic only: arrows show computation or experimental progression, not measured activation flow or traced causal paths.

Recovery performance

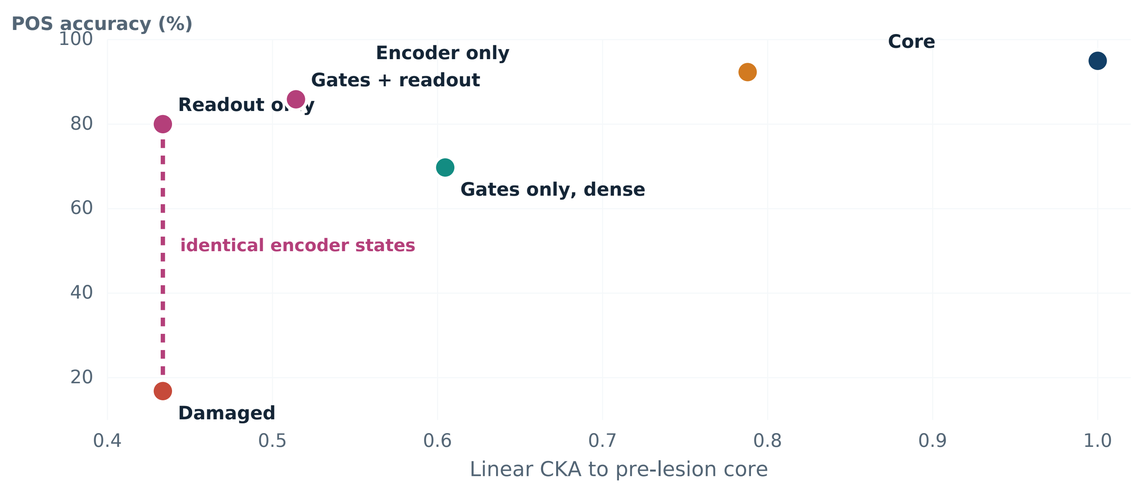
Mean across five seeds



All recovery conditions produced marked improvement over the damaged baseline. Bars show arithmetic means across five full-pipeline runs. Open circles are individual runs; lines show observed ranges, not confidence intervals. The gate-only result is the density-promoting variant.

Accuracy versus representation

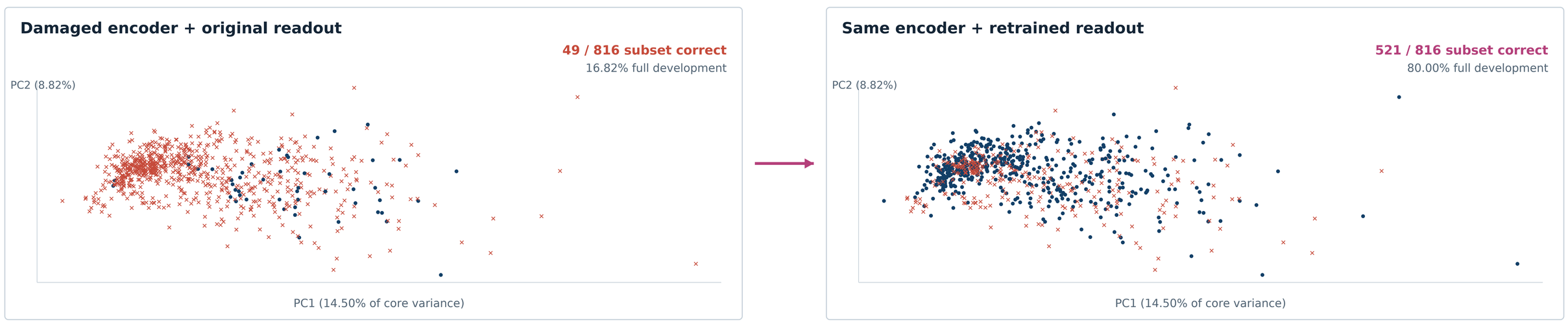
Primary run



Representational analysis showed that behavioral recovery was not necessarily accompanied by restoration of the original hidden-state geometry. CKA measures hidden-state similarity, not recovered circuitry. Readout-only leaves encoder states unchanged yet raises primary-run accuracy from 16.82% to 80.00%.

Readout remapping recovers performance without changing encoder geometry

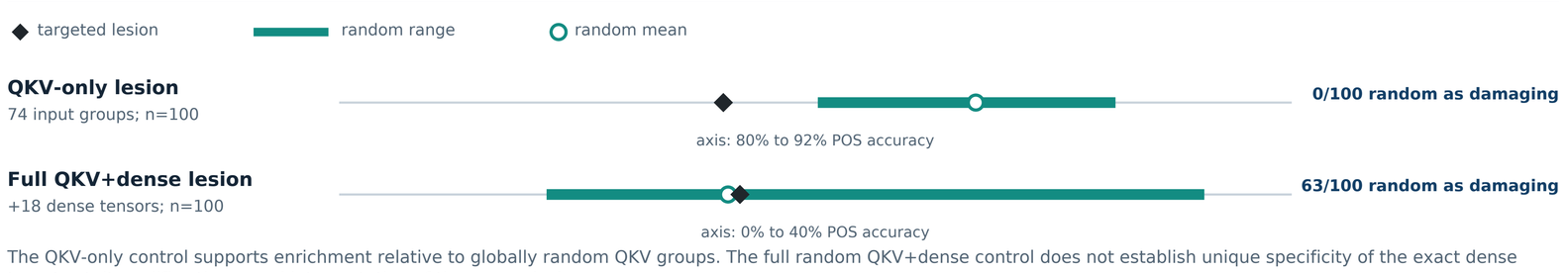
SAME TOKEN REPRESENTATIONS; ONLY THE LINEAR READOUT CHANGES



● correct under shown readout ✕ incorrect under shown readout

The damaged configuration contained task-relevant information that was poorly aligned with the original readout. Both panels show the same 816 POS-balanced token coordinates in one core-fitted PCA basis and shared scale; markers report correctness under each 768-dimensional readout. PC1-PC2 explain 23.32% of core variance and are not the classifier decision space.

Robustness and controls



The QKV-only control supports enrichment relative to globally random QKV groups. The full random QKV+dense control does not establish unique specificity of the exact dense target set; it qualifies the causal interpretation of the severe lesion.

Conclusions

Performance can recover without relearning the underlying transformations.

The extent to which the original geometry is recovered depends on the degrees of freedom of the recovery protocol.

Circuit necessity is conditional on the permitted rerouting strategy. A component may be necessary under the original fixed routing and readout, but cease to be so when the readout can be retrained or surviving transformations can be reselected.

Our evaluations were based on a single task and a single backbone model. The biological relevance of these findings remains unclear.

Methods: Louizos et al. (2018); Cao et al. (2021). Data: Universal Dependencies English-EWT.